HENSELIZATION AND STRICT HENSELIZATION OF A VALUATION RING ARE WEAKLY UNRAMIFIED EXTENSIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 37534-37543 (canonical tag 0ASK) and the counted proof is the proof environment at source lines 37545-37566, which immediately follows it and is its own proof.

Editorial note. The statement and the proof display the chain of inclusions and the map on value groups; both displays are copied exactly as printed, using the macros of the retained preamble.

Editorial note. The four references are one to a result of this chapter, namely the statement about etale extensions of valuation rings, and three to results of other chapters on descriptions of henselization and on colimits of valuation rings; the kinds recorded are the type words appearing in the labels themselves. All four are navigation stubs with their roles recorded in the reasons.

Editorial note. The preamble span runs to line 90, the current end of the retained preamble, which includes the appended environment definitions; the counted text does not use them, and the span is taken whole so that the file version is unambiguous.

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Allowed standard prerequisites (editorial).

- The henselization of a local ring is the filtered colimit of the localizations of etale algebras at primes lying over the maximal ideal, and strict henselization is described by the analogous statement. invoked at proof lines 37545-37549 and again at proof lines 37562-37564. Produces the colimit presentation on which the whole computation rests.
- The localization of an etale algebra over a valuation ring at a prime over the maximal ideal is again a valuation ring, and the map on value groups is bijective. invoked at proof lines 37549-37556. Gives the statement for each term of the colimit, including the bijectivity of the map on value groups.
- A filtered colimit of valuation rings with compatible value groups is a valuation ring, and the value group of the colimit is computed from the colimit of the value groups. invoked at proof lines 37556-37561. Passes from the terms of the colimit to the henselization itself, and then to the strict henselization and the composite inclusion.

Source reference. Original lemma 04GV: sources/stacks-chow/tags.txt, source lines 5846–5846. The reference is the description of the henselization as a colimit of etale

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localizations, used at proof lines 37545-37549; the computation performed on that colimit is inside the counted span.

Source reference. Original lemma 0ASJ: sources/stacks-more-algebra/pinned-original-source.tex, source lines 37480–37480. The reference is applied at proof lines 37549-37556 to each term of the colimit to get a valuation ring with the same value group; it is the input to the colimit passage, which is inside the counted span.

Source reference. Original lemma 0AS4: sources/stacks-chow/tags.txt, source lines 13567–13567. The reference passes the property from the terms to the filtered colimit, used at proof lines 37556-37561; the conclusion about value groups drawn there is inside the counted span.

Source reference. Original lemma 04GW: sources/stacks-chow/tags.txt, source lines 5847–5847. The reference is the strict analogue of the colimit description, used at proof lines 37562-37564 so that the same argument applies to the strict henselization; the identification of the composite inclusion is inside the counted span.

Lemma 0.2. *Let A be a valuation ring. Let A^h, resp. A^{sh} be its henselization, resp. strict henselization. Then*

$$A \subset A^h \subset A^{sh}$$

are extensions of valuation rings which induce bijections on value groups, i.e., which are weakly unramified.

Proof. Write $A^h = \mathrm{colim}(B_i)_{\mathfrak{q}_i}$ where $A \to B_i$ is étale and $\mathfrak{q}_i \subset B_i$ is a prime ideal lying over $\mathfrak{m}_A$, see Algebra, Lemma 04GV. Then Lemma 0ASJ tells us that $(B_i)_{\mathfrak{q}_i}$ is a valuation ring and that the induced map

$$(A \setminus \{0\})/A^* \longrightarrow ((B_i)_{\mathfrak{q}_i} \setminus \{0\})/(B_i)^*_{\mathfrak{q}_i}$$

is bijective. By Algebra, Lemma 0AS4 we conclude that A^h is a valuation ring. It also follows that $(A \setminus \{0\})/A^* \to (A^h \setminus \{0\})/(A^h)^*$ is bijective. This proves the lemma for the inclusion $A \subset A^h$. To prove it for $A \subset A^{sh}$ we can use exactly the same argument except we replace Algebra, Lemma 04GV by Algebra, Lemma 04GW. Since $A^{sh} = (A^h)^{sh}$ we see that this also proves the assertions of the lemma for the inclusion $A^h \subset A^{sh}$. □